

Explainable Federated Learning Framework for Privacy-Preserving Breast Cancer Diagnosis

Mehak Aggarwal¹, Riya Jain²,

¹ IT, Indira Gandhi Delhi Technical University for Women, Delhi, India

² CSEAI, Indira Gandhi Delhi Technical University for Women, Delhi, India

ABSTRACT

Breast cancer is one of the most common and deadly diseases among women worldwide. Early and precise diagnosis is important for improving survival rates. Recent changes in artificial intelligence show great results in supporting proven decision-making through automated disease prediction. This study comes across an explainable federated learning framework for breast cancer diagnosis using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. To protect data privacy while allowing collaborative learning, the dataset was divided into four simulated healthcare clients. Each client trained local machine learning models without sharing raw patient data. The proposed framework combines Federated Logistic Regression, Federated Random Forest, and Federated XGBoost models using a soft ensemble strategy to boost classification performance. Feature standardization and stratified data splitting were done to make sure that the model was adequately trained and validated. Global test split was applied to determine the performance of the federated learning model. Also, techniques like XAI, LIME, and SHAP were utilized for global and local interpretations of model predictions. Experimental results demonstrate that the proposed federated ensemble model can reach high diagnostic accuracy for the distinction of malignant and benign breast tumours with data privacy protection. SHAP and LIME explanations allow clinicians to find the most important diagnostic features that drive the predictions. This approach shows the potential of combining federated learning, ensemble machine learning and explainable AI for developing privacy-preserving and interpretable breast cancer diagnostic systems suitable for real-world healthcare settings.

Keywords: Breast Cancer Diagnosis, Federated Learning, Explainable Artificial Intelligence (XAI), Machine Learning, Ensemble Learning, SHAP, LIME, Privacy-Preserving Healthcare

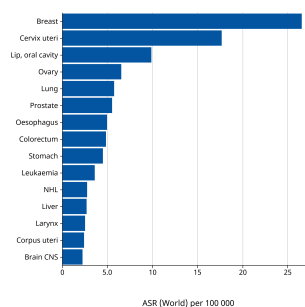
1. INTRODUCTION

Breast cancer is the most frequently diagnosed cancer worldwide, with an estimated 2.3 million new cases diagnosed annually, accounting for nearly 12.5% of all new cancers [11]. It is still one of the causes of cancer death in women. As shown in Fig. 1, breast cancer has the highest age-standardized incidence rate among all major cancer types in India and highlights the increasing burden of breast cancer on the healthcare system. The burden is especially high in India where more than 98,000 women die of breast cancer every year and the latest estimates show that one in two women diagnosed with the disease does not survive. These numbers demonstrate the crucial role of early recognition and proper diagnosis.

1.1 Current Diagnosis Methods

To allow for early diagnosis, clinicians routinely use medical imaging techniques [12]. The performance of these techniques is highly dependent on the expertise and experience of clinicians, leading to variable diagnosis. Therefore, there is an increasing need for intelligent computer-aided decision support systems to help the clinician to make accurate and timely

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
India
(Top 15 cancer sites)



Cancer TODAY | IARC - <https://gco.iarc.who.int/today>
Data version: Globocan 2022 (version 1.1)
© All Rights Reserved 2020

International Agency
for Research on Cancer
World Health
Organization

Figure 1. Age-standardized incidence rates of the fifteen most common cancers in India (GLOBOCAN 2022)

diagnoses.

1.2 AI and Machine learning

To allow for early diagnosis, clinicians routinely use medical imaging techniques [12]. The performance of these techniques is highly dependent on the expertise and experience of clinicians, leading to variable diagnosis. Therefore, there is an increasing need for intelligent computer-aided decision support systems to help the clinician to make accurate and timely

diagnoses.

1.3 Need for Federated Learning

While many traditional machine learning models have achieved impressive predictive performance, they are built upon centralized training methods that involve gathering patient data from numerous healthcare institutions and storing them on a central server. Healthcare records contain highly sensitive patient information and centralised data sharing raises serious concerns around privacy, security and regulation such as HIPAA and GDPR. Such challenges hinder healthcare organizations from collaborating to build reliable diagnostic models. To tackle these challenges, Federated Learning (FL) enables the local devices, e.g., hospital servers or clinical systems, to train models locally and only share the model updates with a central server [1]. This enables multiple healthcare organizations to collectively train a global model without disclosing patient data. However, despite these positives, federated models often work as black-box systems, making their predictions difficult for practitioners to interpret and trust.

1.4 Need for Explainability

Although Federated Learning has many advantages in terms of privacy, the lack of model explainability is a major hurdle for the application of this technique in clinical practice [2]. High accuracy is important, but clinicians must also understand why a model made a particular prediction before incorporating AI-assisted systems into clinical decision-making. To this end, Explainable Artificial Intelligence (XAI) techniques have been developed as an effective way to improve model transparency. Some of the most popular XAI techniques are SHAP and LIME, that provide global and local explanations about the model predictions. However, studies currently focus primarily on predictive performance and do not address explainability within a federated context, despite the increasing interest in merging Federated Learning and XAI for healthcare applications. These negatives highlight the need for a framework that simultaneously ensures high predictive performance, data security and model interpretability.

1.5 Proposed Framework

Driven by the research gaps, this paper proposed an Explainable Federated Learning framework for privacy-preserving breast cancer diagnosis as illustrated in Fig. 2. The proposed framework integrates Federated Learning with Explainable AI to jointly train the model while maintaining patient

privacy and enhancing model transparency. The comparison of different machine learning algorithms is performed for both centralized and federated approaches and SHAP and LIME are used to explain model predictions.

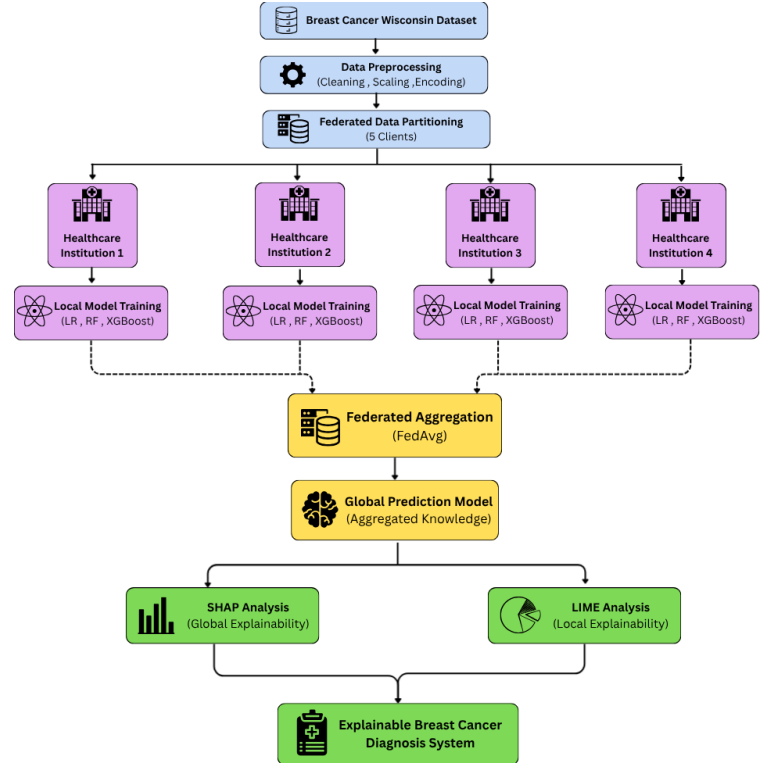


Figure 2. Overall architecture of the proposed explainable federated learning framework

1.6 Contributions

The main contributions of this paper are:

- Design of a Privacy-Preserving Diagnostic Framework based on Explainable Federated Learning.
- Comparative study of different machine learning algorithms with centralized and federated learning approaches
- SHAP and LIME integration for global and local interpretability of model predictions.
- Evaluation of the federated model with client-wise explainability

1.7 Paper Organization

The remainder of the paper is organized as follows. Section II reviews relevant literature, Section III presents the methodology, Section IV presents the experimental results and Section V concludes the paper.

2. LITERATURE REVIEW

The development of privacy-preserving diagnostic frameworks requires a structured evaluation of Federated gathering constraints and post-hoc interpretability methods. This section provides an overview of the current state-of-the-art methods in FL and XAI. It starts with the analysis of individual studies and ends with the fusion of common findings. The section concludes with a discussion of specific research gaps that this work addresses.

2.1 Federated Learning (FL) in Healthcare

Research that seeks to relax data governance and privacy restrictions has profoundly influenced the fundamental patterns of decentralised machine learning.

- **The Centralised Diagnostic Benchmarking study** [4] was based on breast cancer diagnostics from a centralised point of view. They used an optimised implementation of a Support Vector Classifier (SVC) in Python on the structured Breast Cancer Wisconsin Diagnostic Dataset (WBCD). While it reaches good baseline classification metrics, its architectural design points out the inherent anomalies of centralized repositories concerning strict multi-institutional data compliance.
- **The Distributed Diagnostics study** ([1]) explored distributed diagnostics across localised clinical peer group with sophisticated aggregation formulations. They demonstrated that their framework supports cooperative multi-silo training, where independent nodes optimize a shared model without centralising raw datasets. But they noticed that the variations in data distribution directly controls the network speed.

Common Findings: These studies indicate that centralized frameworks [4] achieve strong predictive metrics for structured diagnostic data. However, scaling them to multi-institutional environments needs decentralized FL architectures [1]. Existing FL research often measures performance using complex, serialized models. These models create significant communication overhead among independent client nodes.

2.2 Explainable AI (XAI) in Clinical Pipelines

As predictive models become more complex, explaining them after the fact has become necessary. This change helps turn "black-box" systems into special tools that can be verified.

- **The Interpretability Paradigms study**

[2] explored interpretability paradigms on high-dimensional cancer datasets. It studied post-hoc SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) settings. Our results show that multi-tiered feature explanations can significantly increase expert clinical trust by relating model decisions to real medical features.

- **The Centralised Diagnostic Benchmarking study** ([4]) incorporated game-theoretic feature attributions through SHAP into their centralized SVC pipeline. With this configuration, investigators were able to derive global rankings of features and show clinicians what clinical variables had the highest predictive weight in determining the tumor classification.

Common Findings: The literature suggests that post-hoc explainability frameworks such as SHAP and LIME are very effective in converting mathematical model boundaries to actionable clinical insights [2, 4]. While effective, the majority of the interpretability validations are strictly tied to centralised architectures, and their performance under decentralised, multi-client constraints remains largely unexplored.

2.3 Synthesis of Combined Federated-XAI Frameworks

Recent trends have tried to combine privacy-preserving networks with interpretability layers, resulting in the development of Explainable Federated Learning (XFL) architectures to address the dual challenge of data isolation and opacity.

- The studies on **Distributed Diagnostics** [1] and **Interpretability Paradigms** [2] both show that the technical intersection of distributed aggregation and post-hoc auditing is key to the healthcare adoption, yet the early implementations are still highly segmented. Most combined models lean on heavy deep-learning pipelines that are focused almost exclusively on image modalities, which makes it difficult to deploy them on lightweight tabular frameworks.
- **The Centralised Diagnostic Benchmarking study** ([4]) shows that classical machine learning algorithms are very successful on structured clinical datasets. This structural efficiency, however, has rarely been transferred to a combined privacy-preserving XFL pipeline.

Common Findings: Current combined frameworks across the broader state-of-the-art show that embedding explainability within federated structures does not degrade the classification boundaries intrinsically. However, there is a major opening

for lightweight, communication-efficient machine learning ensemble models applied to structured patient data given the overwhelming dominance of computationally intensive deep learning models in these systems. In summary, the literature indicates that Federated Learning is a promising approach to tackle data privacy issues, and Explainable AI enhances model interpretability. However, few studies combine lightweight ML ensembles with dual explainability techniques in a federated breast cancer diagnosis framework. This observation is the main motivation of the proposed study.

2.4 Research Gap

Critical synthesis of the existing literature reveals several explicit limitations and omissions:

- **Neglect of Explainability:** Most federated healthcare studies focus on optimizing raw predictive accuracy with little consideration for post-hoc model explainability and transparency.
- **Lack of XAI comparative analysis:** There is an extremely limited comparative and simultaneous analysis between SHAP and LIME in federated healthcare settings.
- **Over-Reliance on Heavy Architectures:** The bulk of the existing breast cancer studies are based on heavy deep learning models, while lightweight machine learning ensemble models have received less attention in comparison.
- **Lack of Dual Paradigm Evaluation:** There is a lack of research that has evaluated and bridged the gap between the centralized and federated learning paradigms simultaneously using multiple machine learning models on breast cancer structured datasets.

2.5 Contributions of This Work

To fill these gaps, in this work, we propose a lightweight, communication-efficient and fully explainable diagnostic framework, validated on Breast Cancer Wisconsin Diagnostic Dataset (WBCD), a benchmark structured dataset commonly used to evaluate breast cancer diagnosis models:

- **Multi-Model Centralized & Federated Evaluation:** We systematically evaluate and compare centralized and decentralized learning paradigms using multiple foundational machine learning models: Logistic Regression, Random Forest (RF), and XGBoost.
- **Lightweight Ensemble Architecture:** We propose a communication-efficient Federated Soft Voting Ensemble combining the predictive power of tree-based and linear classifiers over five independent client silos, rather than

using intensive deep neural networks. This configuration provides a good balance between prediction efficiency and communication efficiency.

- **Dual-Layered Federated Auditing (SHAP + LIME):** We address the explainability gap by integrating a complete dual-layer XAI pipeline directly into the federated workflow. This architecture employs SHAP to compute mathematically sound global feature attributions and LIME to provide granular, localized explanations for individual patient profiles, offering clinical transparency and enhancing interpretability without compromising data privacy.

3. METHODOLOGY

3.1 Overview of the Proposed Framework

The overall workflow of the proposed Explainable Federated Learning framework is illustrated in Fig.3. The Breast Cancer Wisconsin (Diagnostic) dataset is preprocessed and split into training and testing sets with an 80:20 train-test split initially. The training data are further split among four simulated healthcare clients to emulate a federated learning environment, while the test set remains at the server to evaluate the global model. Each client trains each Logistic Regression, Random Forest and XGBoost models locally, without sharing any raw patient data. The locally trained models are then aggregated by the Federated Averaging (FedAvg) algorithm to build a global predictive model. Finally, the global test set is used to evaluate the global model, and XAI methods, e.g., SHAP and LIME are used to provide global and local explanations of the model predictions.

3.2 Dataset Description

Table 1. Characteristics of the Breast Cancer Wisconsin (Diagnostic) Dataset

Attribute	Description
Dataset	Breast Cancer Wisconsin (Diagnostic)
Source	UCI Machine Learning Repository
Total Samples	569
Features	30 Numerical Features
Target Classes	Benign (357), Malignant (212)
Missing Values	0

3.3 Data Preprocessing

Before applying the federated training, the Breast Cancer Wisconsin (Diagnostic) dataset was subjected to a preprocessing stage to make sure that the training of the model was unbiased and consistent. The

identifier (ID) attribute was not included as it does not contribute in the prediction of the disease and the diagnosis labels were encoded into binary classes, malignant to 1 and benign to 0. The dataset was then divided into training and testing subsets, with an 80:20 stratified train-test split to maintain the original class distribution. Feature Standardisation: StandardScaler was used. The StandardScaler was fit on the training data only and applied to the global test set, to avoid data leakage. Next, the federated client partitioning and local model training were conducted on the normalised training data.

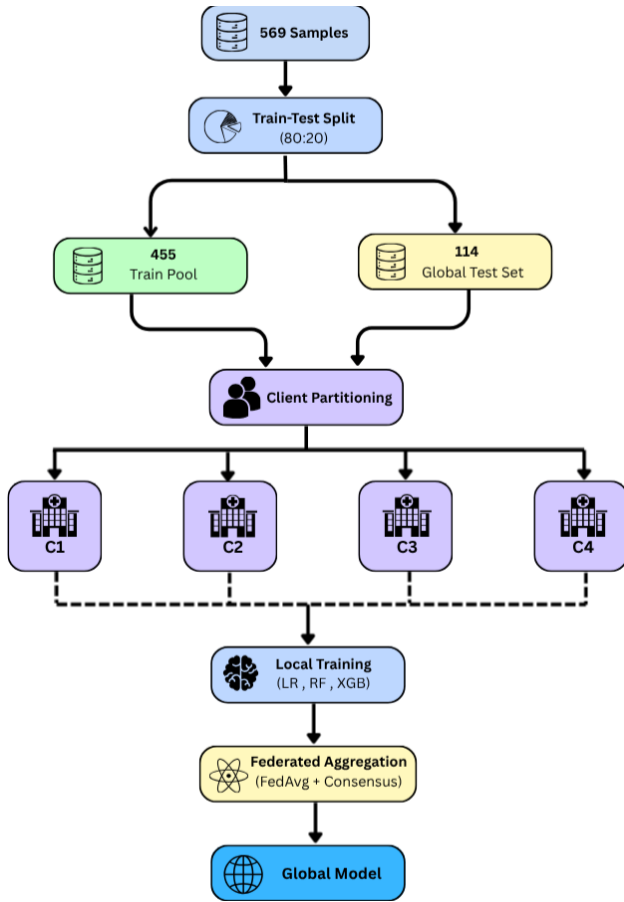


Figure 3. Workflow of the proposed federated learning framework

3.4 Federated Data Partitioning

Table 2. Distribution of training samples among federated clients

Client	Training sample
Client 1	114
Client 2	114
Client 3	114
Client 4	113
Total	455

To mimic a real-world federated healthcare setting,

the preprocessed training dataset was split into four mutually exclusive subsets, each of which corresponds to an independent healthcare institution. The partitioning was performed after preprocessing, to make sure of uniform feature scaling for all clients but without any data leakage from the global test set. Each client trained its own local models using its own data in isolation, without seeing any other client's data. The global test set was kept on the server side and used to evaluate the globally trained model in each training round. The distribution of training samples among the four clients is summarised in Table II.

3.5 Federated Learning Architecture

The client-server federated learning framework consisted of four simulated healthcare institutions and a central aggregation server. After the data partitioning, each client independently trained Logistic Regression, Random Forest and XGBoost models locally on its assigned training data, thereby keeping patient data local during the training phase. After the local training process, the learned model parameters from each client were sent to the central aggregation server where the federated averaging algorithm (FedAvg) was employed to build a global model by aggregating the client updates. In the current experiment, all four clients were used in one round of communication leading to the one-shot federated learning without the iterations between the client-server optimisation process. The performance of the global model was evaluated by testing it against the centralized global test set.

3.6 Machine Learning Models and Ensemble Learning

Three supervised machine learning algorithms, namely Logistic Regression (LR), Random Forest (RF) and Extreme Gradient Boosting (XGBoost), were used as base classifiers for the proposed federated learning framework. The reason of choosing these models is that they provide different learning mechanisms for comparing linear, bagging and boosting learning techniques for breast cancer prediction tasks. All models were trained individually on each federated client's local dataset with the same training settings to ensure a fair comparison of performance. The implementation parameters for each classifier are summarised in Table III.

To further improve the performance of the proposed framework, a Soft Voting Ensemble was constructed using probabilities of Logistic Regression, Random Forest and XGBoost. In contrast to hard voting, soft voting determines the final prediction by taking the

Table 3. Hyperparameter settings of the machine learning models

Model	Hyperparameter Setting
Logistic Regression	max_iter = 1000, random_state = 42
Random Forest	n_estimators = 50, random_state = 42
XGBoost	n_estimators = 50, eval_metric = logloss, random_state = 42

average of the class probabilities produced by each classifier. This led to the creation of an ensemble that exploits the benefits of the individual models and thus yields more reliable and robust breast cancer classification in both centralized and federated scenarios.

3.7 Model Evaluation and Explainability

The proposed framework was evaluated with five popular classification metrics including accuracy, precision, recall, F1-score, and receiver operating characteristic-area under the curve (ROC-AUC). These metrics evaluate the models’ predictive ability by assessing the overall classification performance, reliability of the positive classifications, correctness of the classification of malignant instances and trade-off between precision and recall. Here, Table IV below provides a brief description of the evaluation metrics exploited in this paper.

Table 4. Performance evaluation metrics used in this study

Metric	Purpose
Accuracy	Measures overall classification accuracy
Precision	Evaluates correctness of malignant predictions
Recall	Measures detection capability of malignant cases
F1-Score	Balances precision and recall
ROC-AUC	Measures discriminative performance

Moreover, in the final global model, we included Explainable Artificial Intelligence (XAI) techniques to enhance the transparency and interpretability of the proposed federated learning framework. In this work, we used SHapley Additive exPlanations (SHAP) to determine the importance of each feature, offering global and local explanations of model predictions; Local Interpretable Model-agnostic Explanations (LIME) were employed to provide explanations of each prediction on an instance-specific basis.

By combining these two explainability techniques, we offer a complete interpretation of the model behavior, which will improve the trust in assisted medical decision-making. Table V summarizes the explainability methods used in this paper.

Table 5. Explainability techniques employed in the proposed framework

Technique	Scope	Explanation
SHAP	Global and local	Quantifies feature contribution
LIME	Local	Explains individual predictions

4. RESULTS AND DISCUSSIONS

4.1 Performance Evaluation in a Centralized Setting

The centralized performance of Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) was evaluated using an 80:20 train-test split. The results obtained for Accuracy, Precision, Recall, F1-score, and ROC-AUC are presented in Table VI. As seen

Table 6. Performance of centralized machine learning models

Model	Acc.	Prec.	Rec.	F1	AUC
LR	96.49	97.50	92.86	95.12	0.9960
RF	97.37	100	92.86	96.30	0.9929
XGB	97.37	100	92.86	96.30	0.9940

from Table VI, the highest accuracy (97.37%) was achieved by Random Forest (RF), XGBoost (XGB), whereas Logistic Regression (LR) got the best ROC-AUC (0.996). Overall, the results indicate that all three models demonstrated strong classification performance, establishing reliable baselines for subsequent federated evaluation.

4.2 Cross-Validation Performance Analysis

In order to measure model robustness, stratified five-fold cross-validation was performed. The mean and standard deviation of Accuracy, Precision, Recall, F1-score, and ROC-AUC achieved in cross-validation are shown in Table VII. The cross-validation showed that all models performed well in different folds and for different tools. LR achieved the highest average Accuracy (97.37 ± 1.66) and ROC-AUC (0.9953 ± 0.0053), indicating good generalization, while RF and XGB also maintained competitive performance.

Table 7. Five-fold cross-validation performance of centralized models

Model	Acc.	Prec.	Rec.	F1	AUC
LR	97.37 $\pm$ 1.66	98.63 $\pm$ 1.82	94.36 $\pm$ 5.25	96.33 $\pm$ 2.40	0.9953 $\pm$ 0.0053
RF	95.43 $\pm$ 1.28	94.63 $\pm$ 3.90	93.41 $\pm$ 5.43	93.81 $\pm$ 1.86	0.9885 $\pm$ 0.0083
XGB	95.26 $\pm$ 0.89	94.22 $\pm$ 4.12	93.41 $\pm$ 4.99	93.60 $\pm$ 1.27	0.9935 $\pm$ 0.0040

4.3 Performance Evaluation of Federated Models

The federated approach was tested on four simulated healthcare clients applying FedAvg for aggregating models. The result of the aggregation of Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) models is shown in Table VIII. The federated

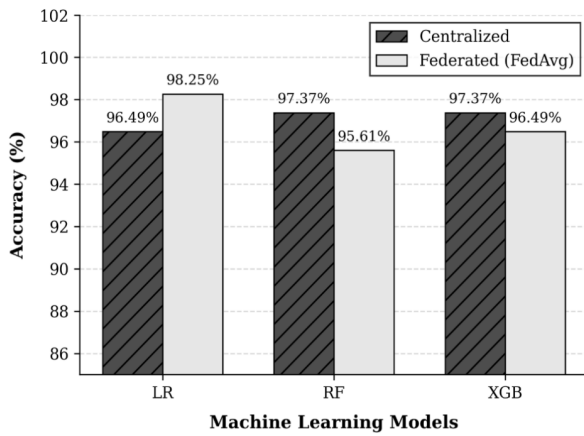
Table 8. Performance of federated global machine learning models

Model	Acc.	Prec.	Rec.	F1	AUC
LR	98.25	100.00	95.24	97.56	0.9980
RF	95.61	97.44	90.48	93.83	0.9947
XGB	96.49	100.00	90.48	95.00	0.9977

models performed excellently in terms of diagnostic metrics without any data exchange between the parties. Federated LR obtained the highest Accuracy (98.25%) and ROC-AUC (0.9980), while RF and XGB also maintained competitive performance, demonstrating the effectiveness of the proposed privacy-preserving mechanism.

4.4 Comparative Analysis of Centralized and Federated Learning

Figure 4 demonstrates the classification accuracy of centralized and federated approaches for the purpose of comparing the impact of decentralized learning. According to Fig. 4, the performance of federated

**Figure 4.** Performance of the proposed federated soft voting ensemble

learning algorithms remained almost at the same level compared to the centralized approaches. Although

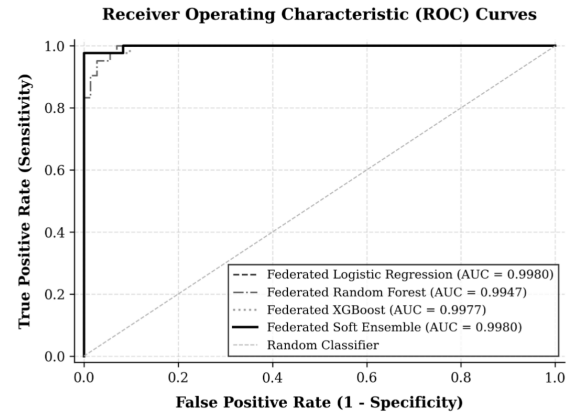
Federated LR provided better accuracy, RF and XGB demonstrated a slight decrease in accuracy, thus showing that privacy preservation is possible without significant loss of prediction performance

4.5 Performance of the Proposed Soft Voting Ensemble

A Federated Soft Voting Ensemble was constructed using LR, RF, and XGB to improve prediction robustness. The evaluation results are presented in Table X and Fig. 5. The proposed Soft Voting

Table 9. Performance of the Proposed Federated Soft Voting Ensemble

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
FSVE	97.37	97.00	97.00	97.00	0.9980

**Figure 5.** ROC curves of federated models and the proposed soft voting ensemble

Ensemble achieved 97.37% Accuracy and an ROC-AUC of 0.9980, demonstrating robust and reliable diagnostic performance. The ROC curves in Fig. 5 further confirm the excellent discriminative capability of the proposed ensemble.

4.6 Explainability Analysis

Although the proposed federated ensemble achieved high predictive performance, understanding the rationale behind its predictions is equally important for clinical decision-making. In order to improve the explainability of the federated ensemble, SHAP and LIME methods were applied to provide a

global and local explanation, respectively. While the SHAP method provides information about feature importance across all data instances, the LIME method gives an explanation of the prediction for specific patients, thus providing two different perspectives on model behavior.

4.6.1 Global Feature Interpretation using SHAP

Fig. 6 presents the SHAP summary plot illustrating the global importance of diagnostic features. The higher the value of SHAP is, the more influence this feature has on the model's prediction. As shown in

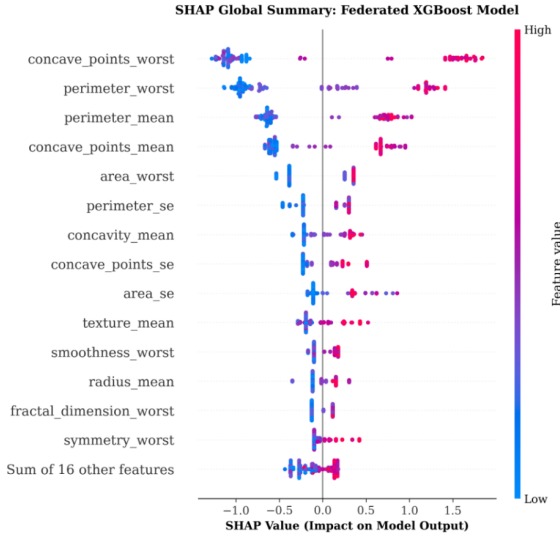


Figure 6. SHAP summary plot showing the global importance of diagnostic features

Figure 6, the SHAP summary plot is used to visualize the most influential diagnostic variables that help classify breast cancer. In turn, SHAP enables a clear global interpretation of the suggested model.

4.6.2 Local Prediction Interpretation using LIME

Figure 7 shows the LIME explanation of individual predictions based on identifying the features having the highest impact on each decision, thereby making it easier to interpret AI-assisted clinical decisions. In contrast to SHAP, LIME offers an individual explanation of a diagnosis, thus enhancing its transparency. LIME explains individual predictions by highlighting the features that most strongly influence each classification, thereby improving the interpretability of AI-assisted clinical decisions.

4.7 Overall Discussion

The experimental results show that the developed Explainable Federated Learning framework successfully establishes a tradeoff between model

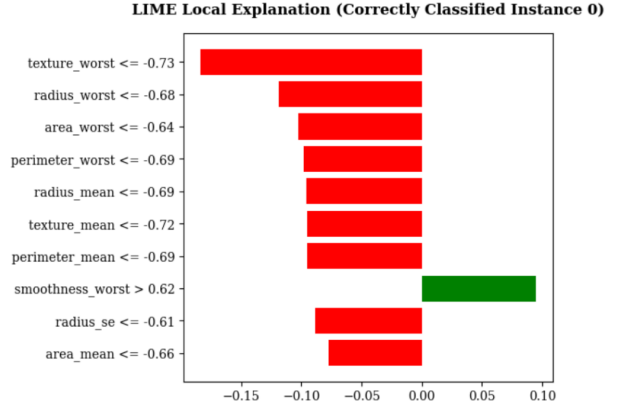


Figure 7. LIME explanation for a representative breast cancer prediction

prediction capability, privacy-preserving, and interpretability. The experimental results show that the proposed Explainable Federated Learning framework successfully strikes a balance between model prediction capability, privacy-preserving, and interpretability. The federated learning framework managed to sustain a good level of diagnostic capability without exchanging any private data of the patients, while the application of SHAP and LIME contributed to model interpretability by generating global and local explanations. Besides, the developed Soft Voting Ensemble provided high classification accuracy and ROC-AUC, which means higher reliability. The application of SHAP and LIME increased the interpretability of the model by generating global and local explanations.

5. CONCLUSION AND FUTURE WORK

In this paper, we developed an Explainable Federated Learning framework that could provide privacy-preserving breast cancer diagnosis using the Breast Cancer Wisconsin (Diagnostic) dataset. The combination of Federated Learning with Logistic Regression, Random Forest, XGBoost and Soft Voting Ensemble enabled us to build highly accurate diagnostic models without sharing patients' sensitive information among healthcare organizations. The experimental study showed that the federated models had high classification accuracy, and our proposed ensemble had an outstanding diagnostic performance with an ROC-AUC score of 0.9980. Additionally, SHAP and LIME were used to enable global and local explanations, which increased the explainability and interpretability of our proposed diagnostic solution. Even though we tested the developed framework on simulated healthcare clients and one benchmark dataset, the achieved results proved its capabilities for

privacy-preserving and explainable clinical decision support. Further research will be related to validation of our framework on multi-institutional datasets, investigation of non-IID distribution of client data, development of federated deep learning models and evaluation of our approach in real healthcare environment.

6. REFERENCES

1. A. Gayen, R. Das, A. Mondal, S. Singha, and A. Jana, "A Survey: Federated Learning in Healthcare," *Computers & Electrical Engineering*, vol. 135, Art. no. 111195, Jul. 2026, doi: 10.1016/j.compeleceng.2026.111195.
2. A. Chaddad, J. Peng, J. Xu, and A. Bouridane, "Survey of Explainable AI Techniques in Healthcare," *Sensors*, vol. 23, no. 2, Art. no. 634, Jan. 2023, doi: 10.3390/s23020634.
3. D. Kumar, C. Verma, and Z. Illés, "Federated Learning with Explainable AI for Liver Disease Prediction: A Privacy-Preserving Approach," *Intelligence-Based Medicine*, vol. 12, Art. no. 100285, 2025, doi: 10.1016/j.ibmed.2025.100285.
4. T. A. Sathi, R. Jany, A. Azad, S. A. Alyami, N. Alotaibi, I. Hussain, and M. A. Hossain, "Explainable Federated Learning for Multi-Class Heart Disease Diagnosis via ECG Fiducial Features," *Diagnostics*, vol. 15, no. 24, Art. no. 3110, 2025, doi: 10.3390/diagnostics15243110.
5. L. Zhao, H. Xie, L. Zhong, and Y. Wang, "Explainable Federated Learning Scheme for Secure Healthcare Data Sharing," *Health Information Science and Systems*, vol. 12, Art. no. 49, 2024, doi: 10.1007/s13755-024-00306-6.
6. P. Ducange, F. Marcelloni, A. Renda, and F. Ruffini, "Federated Learning of XAI Models in Healthcare: A Case Study on Parkinson's Disease," *Cognitive Computation*, vol. 16, pp. 3051–3076, 2024, doi: 10.1007/s12559-024-10332-x.
7. A. M. Salih, Z. Raisi-Estabragh, I. Boscolo Galazzo, P. Radeva, S. E. Petersen, K. Lekadir, and G. Menegaz, "A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME," arXiv preprint arXiv:2305.02012, 2024.
8. T. Tunduny and B. Shibwabo, "Explainable AI Approaches in Federated Learning: Systematic Review," *JMIR AI*, vol. 5, e69985, 2026, doi: 10.2196/69985.
9. E. Briola, C. C. Nikolaidis, V. Perifanis, N. Pavlidis, and P. S. Efraimidis, "A Federated Explainable AI Model for Breast Cancer Classification," in *Proceedings of the European Interdisciplinary Cybersecurity Conference (EICC 2024)*, Xanthi, Greece, Jun. 5–6, 2024, pp. 194–201, doi: 10.1145/3655693.3660255.
10. B. Karthiga, K. R. Praneeth, V. Saravanan, and T. K. Rama Krishna Rao, "Enhancing Cancer Detection in Medical Imaging Through Federated Learning and Explainable Artificial Intelligence: A Hybrid Approach for Optimized Diagnostics," *Egyptian Informatics Journal*, vol. 31, Art. no. 100751, 2025, doi: 10.1016/j.eij.2025.100751.
11. A. M. Al-Hejri, A. H. Sable, R. M. Al-Tam, M. A. Al-antari, S. S. Alshamrani, K. M. Alshmrany, and W. Alatebi, "A Hybrid Explainable Federated-Based Vision Transformer Framework for Breast Cancer Prediction via Risk Factors," *Scientific Reports*, vol. 15, Art. no. 18453, 2025, doi: 10.1038/s41598-025-96527-0.
12. G. Pratap and R. S. Sirisati, "Enhanced Breast Cancer Detection Using Federated Learning With Explainable AI and Multimodal Data Integration," *International Journal of Advanced Signal and Image Sciences*, vol. 11, no. 6s, pp. 1211–1224, 2025.
13. S. Biswas, S. Saha, M. S. Uddin, and R. Mostafiz, "An Explainable and Federated Deep Learning Framework for Skin Cancer Diagnosis," *PLOS ONE*, vol. 20, no. 7, e0324393, 2025, doi: 10.1371/journal.pone.0324393.
14. N. Aibe, K. Karasawa, M. Aoki, K. Akahane, Y. Ogawa, E. Ogo, S. Kanamori, J. Kawamori, A. I. Saito, K. Shiraishi, H. Sekine, S. Tachiiri, M. Yoshimura, and C. Yamauchi, "Results of a Nationwide Survey on Japanese Clinical Practice in Breast-Conserving Radiotherapy for Breast Cancer," *Journal of Radiation Research*, vol. 60, no. 1, pp. 142–149, 2019, doi: 10.1093/jrr/rry095.
15. V. Anderson, P. Pramanik, and H. K. Robinson, "Obesity and Sociodemographic Factors in Luminal Breast Cancer," arXiv preprint arXiv:2605.30591, 2026.
16. A. S. Thakur and A. Sarkar, "Seeing Below the Limit of Detection: A Censored-Poisson Bayesian Latent-Growth Change-Point Detector (the Span Detector) for Serial ctDNA in HR+/HER2 Metastatic Breast Cancer," arXiv preprint arXiv:2606.11876, 2026.
17. X. Feng, L. Song, S. Wang, H. Song, H. Chen, Y. Liu, C. Lou, J. Zhao, Q. Liu, Y. Liu, R. Zhao, K. Xing, S. Li, Y. Yu, Z. Liu, C. Yin, B. Han, Y. Du, R. Xin, L. Huang, Z. Fan, and F. Zhou, "Accurate Prediction of Neoadjuvant Chemotherapy Pathological Complete Remission

-
- (pCR) for the Four Sub-Types of Breast Cancer,” *IEEE Access*, vol. 7, pp. 134697–134706, 2019, doi: 10.1109/ACCESS.2019.2941543.
18. H. Q. Vo, L. Wang, K. K. Wong, C. F. Ezeana, X. Yu, W. Yang, J. Chang, H. V. Nguyen, and S. T. C. Wong, “Frozen Large-Scale Pretrained Vision-Language Models are the Effective Foundational Backbone for Multimodal Breast Cancer Prediction,” *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 5, pp. 3234–3246, 2025, doi: 10.1109/JBHI.2024.3507638.
 19. P. Avhad, V. Kshirsagar, M. Nakhua, and U. Ranjan, “Devising a Solution to the Problems of Cancer Awareness in Telangana,” arXiv preprint arXiv:2506.21500, 2025.
 20. Z. Su, Y. Guo, R. Wesolowski, G. Tozbikian, N. S. O’Connell, M. K. K. Niazi, and M. N. Gurcan, “Computational Pathology for Accurate Prediction of Breast Cancer Recurrence: Development and Validation of a Deep Learning-Based Tool,” 2025.
 21. H. Khan, Z. Su, H. Zhang, Y. Wang, B. Ning, S. Wei, H. Guo, Z. Li, and M. K. K. Niazi, “Predicting Neoadjuvant Chemotherapy Response in Triple-Negative Breast Cancer Using Pre-Treatment Histopathologic Images,” *Cancers*, vol. 17, 2025.
 22. M. Al Hasani, “Feature Dimensionality Outweighs Model Complexity in Breast Cancer Subtype Classification Using TCGA-BRCA Gene Expression Data,” arXiv preprint arXiv:2605.06562, 2026.
 23. A. Ghasemi, S. Hashtarkhani, D. L. Schwartz, and A. Shaban-Nejad, “Explainable Artificial Intelligence in Breast Cancer Detection and Risk Prediction: A Systematic Scoping Review,” *Cancer Innovation*, vol. 3, p. e136, 2024, doi: 10.1002/cai2.136.
 24. S. Bai, S. Nasir, R. A. Khan, A. Meyer, and H. Konik, “Breast Cancer Diagnosis: A Comprehensive Exploration of Explainable Artificial Intelligence (XAI) Techniques,” arXiv preprint arXiv:2406.00532, 2025.